EXPERIMENTAL ASSUMPTIONS - NOT FOR CONSTRUCTION, FABRICATION OR ORDERS

Experimental basis and selected scenarios

England / UK National Annex basis / published UK supplier data

What this issue does: compares four local first-floor joist bays, three timber headers in a proposed bearing stud wall, two steel support hypotheses, six generic cavity-lintel cases and basement load/quantity sensitivities. It appends the current geometry coordination drawings and retains links to the complete 572-element architectural register. It is not a whole-building structural design.

8.59 m is the east-west bounding width of a fragmented floor mesh, NOT a verified clear joist span. No central spine beam is proposed. Direction is assessed by local bay and supports below.

Low, reference and high are chosen sensitivity cases, not measured site bounds, probabilities or guaranteed worst cases. Conditions outside these cases remain possible.

Input / units	Lower	Reference	Higher
Floor permanent Gk / kN/m2	0.75	1.00	1.35
Floor imposed Qk / kN/m2	1.50	1.50	1.50
Added span allowance / m	0.10	0.15	0.25
Garage-head permanent line load / kN/m	4.0	7.0	10.0
Extra permanent midspan point load / kN	0.0	0.0	10.0
Assumed K0 / dimensionless	0.40	0.50	0.70
Surface surcharge / kPa	5.0	10.0	20.0
Water head above base underside / m	0.00	1.50	3.05

Gk includes an assumed 0.35 kN/m2 lightweight-partition allowance; remaining floor/deck/finish/ceiling/services allowance is 0.40 / 0.65 / 1.00 kN/m2. These are budgets, not measured build-ups. Stone, screed, masonry partitions, baths and heavy storage can exceed them. Qk=1.5 kN/m2 domestic floor area loading; supplier table also includes a separate 2 kN concentrated imposed-load check.

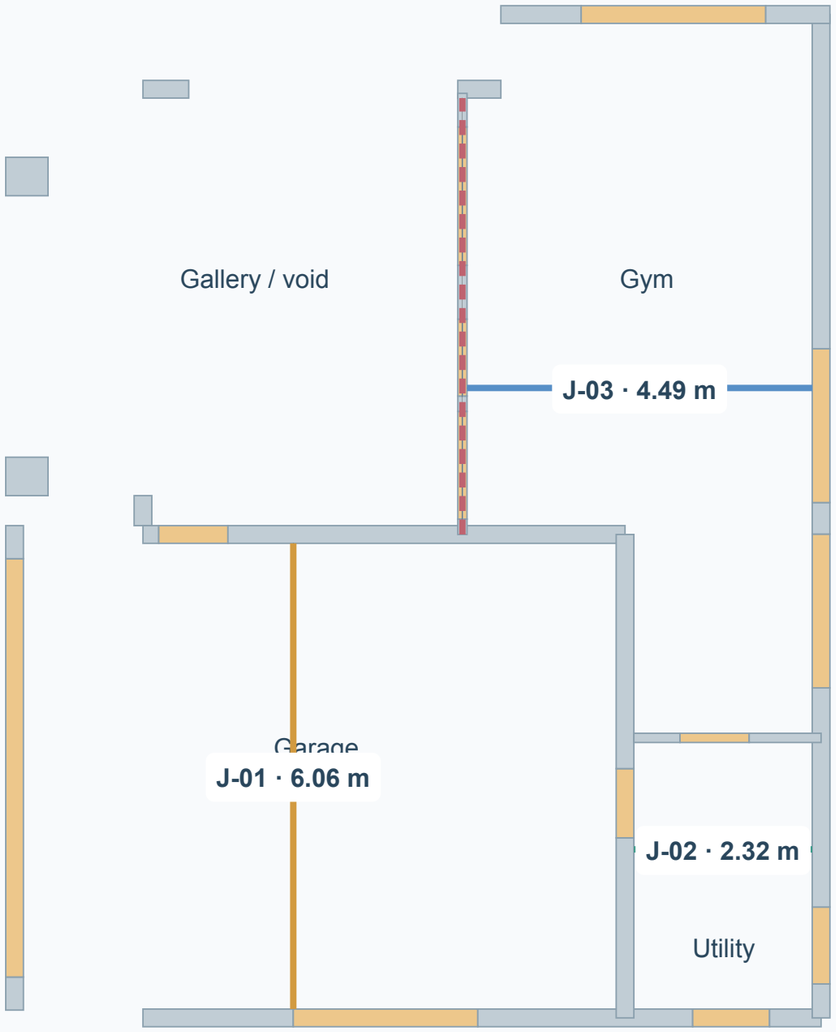
Owner decisions: brick outer leaf, insulated cavity and load-bearing blockwork for new external house walls; a load-bearing timber stud wall is the preferred gym partition option. Small framing/layout adjustments are acceptable. No ground investigation or structural engineer report is available.

EXPERIMENTAL ASSUMPTIONS - NOT FOR CONSTRUCTION, FABRICATION OR ORDERS

Local spans and the supporting walls below

First floor / correcting the whole-wing bounding-box assumption

First-floor span directions / ground-floor supports below



Corrected span basis

8.59 m is the overall floor-mesh width.
It crosses separate rooms and a dividing wall.
It is not a clear joist span.

J-01 / garage

6.06 m north-south candidate.
Parallel to the large west garage opening.
South and garage/store bearing lines unverified.

J-02 / utility

2.32 m east-west candidate.
Garage dividing wall to east external wall.

J-03 / gym

4.49 m east-west candidate.
Dashed red line: proposed bearing timber stud wall.
Timber headers and supporting studs at openings.

J-04 / west wing (outside this diagram)

4.95 m east-west candidate.
Check original-house junction and its openings.

MODEL DIMENSIONS ONLY · BAY ARROWS, NOT A JOIST LAYOUT · NOT FOR CONSTRUCTION

EXPERIMENTAL ASSUMPTIONS - NOT FOR CONSTRUCTION, FABRICATION OR ORDERS

First-floor joist supplier comparisons

James Jones JJI Table 16 / conditional choices at 400 mm centres

Bay	Scenario	Engineering span	First shallow candidate	Table span limit	Build-up*	SLS end reaction / m
J-01 / First floor over garage	low	6.156 m	JJI 300D	6.286 m	357 mm	6.93 kN/m
J-01 / First floor over garage	reference	6.206 m	JJI 300D	6.286 m	357 mm	7.76 kN/m
J-01 / First floor over garage	high	6.306 m	JJI 350D	6.817 m	407 mm	8.99 kN/m
J-02 / First floor over utility	low	2.420 m	JJI 195A+	4.024 m	252 mm	2.72 kN/m
J-02 / First floor over utility	reference	2.470 m	JJI 195A+	3.621 m	252 mm	3.09 kN/m
J-02 / First floor over utility	high	2.570 m	JJI 195A+	3.621 m	252 mm	3.66 kN/m
J-03 / First floor over northern gym	low	4.590 m	JJI 220B+	4.724 m	277 mm	5.16 kN/m
J-03 / First floor over northern gym	reference	4.640 m	JJI 220C	4.669 m	277 mm	5.80 kN/m
J-03 / First floor over northern gym	high	4.740 m	JJI 220D	5.182 m	277 mm	6.75 kN/m
J-04 / West wing first floor	low	5.050 m	JJI 220D	5.297 m	277 mm	5.68 kN/m
J-04 / West wing first floor	reference	5.100 m	JJI 220D	5.182 m	277 mm	6.38 kN/m
J-04 / West wing first floor	high	5.200 m	JJI 235D	5.462 m	292 mm	7.41 kN/m

*Build-up = joist depth + 22 mm deck + 15 mm ceiling + 20 mm finish. Not a complete acoustic/fire/services specification. Existing visual floor envelope is commonly 200 mm. The floor levels, stair headroom and supporting heads must be recoordinated.

James Jones Table 16, domestic intermediate floors, service class 1, simply supported, adequately restrained with 22 mm chipboard and 15 mm plasterboard; engineering span, not cut length. Uses 1.35 kN/m2 table conservatively for reference Gk=1.0, without interpolation. Published vibration assumptions include a 4 m floor width; actual room widths and finishes require supplier design. Bearings, holes, hangers, trimming, fire and acoustics remain unverified.

The 2.0 kN/m2 permanent + 2.0 kN/m2 imposed stress case is outside this table. No choice is returned. Full 480/600 mm alternatives and source data are retained in trials.json. Supplier-table vibration assumptions do not establish performance in these actual rooms.

EXPERIMENTAL ASSUMPTIONS - NOT FOR CONSTRUCTION, FABRICATION OR ORDERS

Preferred gym wall: load-bearing timber studs

Normal structural timber framing can retain the glazed openings; no masonry piers assumed

Upper sensitivity geometry: half 4.49 m gym bay plus half 5.53 m gallery bay = 5.01 m tributary width. Assumes one domestic floor, including the scenario partition budget. Gallery voids may reduce load but are not deducted. Roof/loft/upper masonry and concentrated reactions are NOT automatically included. Headers share load equally between two side-by-side plies, requiring designed connections and bearing to both.

Opening	Scenario	Conditional header	ULS end reaction	Direct bearing indication	Final deflection	Extra head height*
TH-01 South glazing / 1.40 m	low	2 x 45 x 145 mm C24	12.31 kN	89 mm	4.48 mm	35 mm
TH-01 South glazing / 1.40 m	reference	2 x 45 x 195 mm C24	14.06 kN	102 mm	2.59 mm	85 mm
TH-01 South glazing / 1.40 m	high	Outside C24 shortlist	-	-	-	-
TH-02 Doorway / 1.00 m	low	2 x 45 x 145 mm C24	9.03 kN	65 mm	1.44 mm	0 mm
TH-02 Doorway / 1.00 m	reference	2 x 45 x 145 mm C24	10.41 kN	75 mm	1.91 mm	0 mm
TH-02 Doorway / 1.00 m	high	2 x 45 x 220 mm C24	19.57 kN	141 mm	2.24 mm	43 mm
TH-03 North glazing / 1.80 m	low	2 x 45 x 195 mm C24	15.62 kN	113 mm	4.83 mm	85 mm
TH-03 North glazing / 1.80 m	reference	2 x 45 x 220 mm C24	17.69 kN	128 mm	4.36 mm	110 mm
TH-03 North glazing / 1.80 m	high	Outside C24 shortlist	-	-	-	-

*Header plus an illustrative 90 mm top-plate allowance, compared with current 200 mm window head / 267 mm door head zones. This is not a plate detail. The selected header may not fit without lowering glazing, integrating the binder, changing the header type or revising floor depth.

Current heads start at 2.40 m (windows) and 2.333 m (door), with floor underside at 2.60 m: only 200 / 267 / 200 mm head zones. A header, top plates and bearing arrangement must fit together; any required lowering of glazing or floor-depth change must be shown. Adjacent 200 mm solid strips may need minor widening for stud packs. Preserve doorway clear width; revise glazing only as needed.

Not checked: stud axial/buckling capacity and spacing, plate compression, connections, ply load sharing, whole-wall stability, fire and acoustic performance. Direct bearing length is an indication using kc90=1; it is not a stud-pack specification. Higher cases outside the C24 shortlist require revised framing or UK-supplied LVL design.

Below the wall: The model has a full basement under this wall. The ground-floor slab is an architectural envelope, not an established support for the stud wall. A designed transfer/rib/wall/beam and foundation load path is needed below; changing to timber does not remove this requirement.

EXPERIMENTAL ASSUMPTIONS - NOT FOR CONSTRUCTION, FABRICATION OR ORDERS

Steel screening and the limits of the result

Garage-door head / full-span gym alternative retained for comparison only

Study	Scenario	Conditional S355 UB	Engineering span	ULS moment	ULS reaction / end	SLS deflection
B-01	low	203x133x30 UB	5.540 m	22.24 kNm	16.06 kN	8.66 mm
B-01	reference	305x165x40 UB	5.590 m	39.00 kNm	27.90 kN	5.27 mm
B-01	high	356x171x51 UB	5.690 m	76.57 kNm	47.08 kN	6.07 mm
B-02	low	254x146x31 UB	5.775 m	32.25 kNm	22.34 kN	8.37 mm
B-02	reference	254x146x31 UB	5.825 m	36.03 kNm	24.74 kN	9.57 mm
B-02	high	305x165x40 UB	5.925 m	62.46 kNm	35.42 kN	8.53 mm

B-01 / West garage door head
Garage joists in the north-south candidate run parallel to this opening, so no distributed floor tributary width is assigned here. Assumed wall/roof-equivalent permanent line load 4 / 7 / 10 kN/m is a sensitivity only. Higher case adds a 10 kN permanent midspan point load. The actual roof, upper facade and edge-floor reactions remain unknown.

B-02 / Gym west full-span steel alternative (not preferred)
Comparison only if the timber stud-wall route cannot be made to work. One gym floor bay only: 4.49 / 2 = 2.245 m tributary width. Assumes simple end supports; neither is established. No gallery, loft or roof reaction included. Higher case adds 10 kN permanent midspan load. Not the preferred scheme.

Six-section British Steel S355 UB shortlist. Simple span, major-axis bending, symmetric UDL plus central permanent point load, no axial load/torsion, no holes and effective lateral/torsional restraint assumed. Elastic bending and web shear screen only, not a member approval. Connections, lateral-torsional buckling, patch loading, bearings, robustness, fire and support capacities are not checked. The stated section may be inadequate once these are considered.

Chosen trial criteria: total characteristic deflection <= min(L/360,10 mm), imposed-only <= L/360. This is a project screening criterion, not a universal regulatory limit. Glazing/masonry may need tighter limits, camber and differential-movement checks.

ULS = 1.35G + 1.5Q (one adverse variable action). M=wL^2/8+PL/4; R=wL/2+P/2; delta=5wL^4/(384EI)+PL^3/(48EI). E=210000 N/mm^2. Supplier I cm4 x10000; Wel cm3 x1000. Self weight included per candidate.

Bending uses elastic section modulus and class <=3 eligibility under pure bending. Shear uses the flat-web area conservatively, rejects high-shear interaction and applies a slenderness gate. Neither screen verifies lateral-torsional restraint/buckling, axial effects, load introduction or support capacity. All candidate results, including failures, are in trials.json.

EXPERIMENTAL ASSUMPTIONS - NOT FOR CONSTRUCTION, FABRICATION OR ORDERS

Opening-head sensitivity and register coverage

103 architectural heads registered / 6 generic external cavity-wall comparisons

Reference / modelled head	Proxy clear span	CG150/100 length	SWL	Lower load	Reference load	Higher load
O-234CDFC / New wing south head 0	2.40 m	2700 mm	26 kN	5.90 kN	23.62 kN	82.66 kN / ratio outside table
O-136CB09 / New wing south head 1	1.00 m	1350 mm	15 kN	2.46 kN	9.84 kN	34.44 kN / ratio outside table
O-8E24858 / New wing east head 0	1.00 m	1350 mm	15 kN	2.46 kN	9.84 kN	34.44 kN / ratio outside table
O-C883714 / New wing east head 1	2.00 m	2400 mm	22 kN	4.92 kN	19.68 kN	68.88 kN / ratio outside table
O-FC3BFB0 / New wing east head 2	2.00 m	2400 mm	22 kN	4.92 kN	19.68 kN	68.88 kN / ratio outside table
O-3037477 / North ground facade east head 0	2.40 m	2700 mm	26 kN	5.90 kN	23.62 kN	82.66 kN / ratio outside table

Modelled head extent is only a proxy. Catalogue length comparison adds 150 mm bearing at each end, but neither actual clear opening nor bearing length/masonry strength has been verified. Do not place an order from this comparison.

Generic masonry sensitivity: unit weight 20 kN/m3, outer leaf 102.5 mm, loaded masonry heights 0.6 / 1.2 / 2.4 m and inner:outer load ratios 1:1 / 3:1 / 6:1. No arching reduction or point loads. The first two ratios are explicitly listed by the supplier; 6:1 is outside this data. A load greater than the stated SWL fails even this limited comparison.

Catnic SWL is total unfactored UDL, not an ultimate resistance. The comparison uses service loads. Catalogue cavity is 150-165 mm; inner leaf 100-115 mm; outer leaf is stated as 100-102 mm. Confirm suitability for the chosen 102.5 mm brick and actual wall build-up with the supplier.

The other 97 heads are not automatically sized: some are internal or retained, others lack a reconciled cavity/opening/load basis. The separate timber-wall header study and garage-head steel hypothesis do not turn the architectural head register into a complete engineered lintel schedule.

EXPERIMENTAL ASSUMPTIONS - NOT FOR CONSTRUCTION, FABRICATION OR ORDERS

Basement earth pressure and groundwater sensitivity

115.525 m2 architectural footprint / 3.05 m retained-height illustration

Scenario	K0	Soil dry / sat. kN/m3	Surcharge kPa	Water head m	Base pressure kPa	Resultant kN/m	Base moment kNm/m	Buoyancy kN
low	0.40	18.00 / 20.00	5.00	0.00	23.96	39.59	43.35	0
reference	0.50	19.00 / 21.00	10.00	1.50	42.83	66.08	71.50	1700
high	0.70	20.00 / 22.00	20.00	3.05	69.95	128.02	151.86	3457

3.05 m retained-height illustration from ground datum to underside of current 250 mm base. Effective-stress earth pressure with assumed K0 plus hydrostatic water and uniform surface surcharge. Cantilever base moment is a comparison model, not the designed restraint condition. No beneficial drainage assumed. Adjacent foundations, compaction, heave, swelling, accidental surcharge and flooding above grade are outside these scenarios.

p(z)=K0[q+gamma_dry min(z,zw)+(gamma_sat-9.81)max(z-zw,0)]+9.81max(z-zw,0). Integrate p for force and p(H-z) for cantilever base moment. No geotechnical design factors or resistance check.

Unfactored uplift=9.81 x water head above base underside x plan area. Base self weight only=25 x area x thickness; not full building flotation resistance.

Current 250 mm base self-weight alone is 722 kN. It excludes walls, roof slab and superstructure; no full-building flotation safety factor or pass/fail is calculated. Groundwater at or above grade, blocked drains and construction stages require separate assessment.

Not established: soil stratigraphy, effective stress parameters, drainage conditions, clay heave/shrinkage, ground bearing capacity, settlement, adjoining foundation surcharge, wall restraint, excavation sequence, anchorage or structural continuity. Values can fall outside all three chosen cases.

No reinforcement or foundation dimensions issued. Concrete bending/shear/punching, crack control, joints, waterstops, waterproofing coordination, flotation design and temporary works require a site-specific structural/geotechnical scheme.

EXPERIMENTAL ASSUMPTIONS - NOT FOR CONSTRUCTION, FABRICATION OR ORDERS

Concrete volumes and masonry quantity rates

Geometric sensitivities / UK module sizes / no procurement quantities

Illustrative thickness	Base slab only	Wall shell only, 2.58 m high
230 mm	26.57 m3	25.95 m3
250 mm	28.88 m3	28.16 m3
300 mm	34.66 m3	33.63 m3
350 mm	40.43 m3	39.06 m3
400 mm	46.21 m3	44.43 m3

Same stepped outside footprint for each case. Wall volume = (outside polygon area - inward-offset polygon area) x 2.58 m, counting corners once. No openings deducted. Slab and wall thicknesses are independent geometry cases, not paired engineering selections. Excludes basement roof slab, footings/piles, blinding, reinforcement, waste and construction allowances.

Per 100 m2 NET leaf face	No waste	5% allowance	10% allowance
215 x 102.5 x 65 mm brick	5926	6223	6519
440 x 215 mm face block	988	1038	1087

Using 10 mm nominal joints: 1/(0.225 x 0.075) = 59.26 bricks/m2; 1/(0.450 x 0.225) = 9.88 blocks/m2. Counts are rounded up for the stated 100 m2 example. Actual bond, returns, specials and pack sizes alter the order.

Modular arithmetic for one leaf, per 100 m2 NET wall face. Not a whole-house order. Returns, openings, bond, specials and pack rounding excluded. Block strength/mortar/ties not designed.

Whole-house brick and block totals remain unissued: the model uses a 230 mm architectural wall shell and decorative surface fragments, not a reconciled brick/cavity/block assembly. Summing those fragments would misstate the order. Establish net leaf areas, actual openings, reveals and the final wall thickness first.

EXPERIMENTAL ASSUMPTIONS - NOT FOR CONSTRUCTION, FABRICATION OR ORDERS

Calculation audit, references and next design gates

Source-linked formulas / reusable input and result files / UK-T01

Trial two-ply C24: b=90 mm, h=145/195/220 mm, fm,k=24, fv,k=4, fc90,k=2.5 N/mm^2, Emean=11000, Gmean=690 N/mm^2. gammaM=1.3; kmod=.6 permanent and .8 medium-term, both checked; kh=ksys=1, kcr=.67, full end shear. SC1 kdef=.6, domestic psi2=.3. Bending plus shear deformation; final=deltaG(1+.6)+deltaQ(1+.3x.6). Direct bearing indication uses kc90=1 and no effective-length enhancement. Inst <= min(L/300,8mm), final <= min(L/250,10mm) are chosen screening limits. Equal load/bearing to both restrained plies assumed; connections, studs, stability and whole-wall design outstanding.

UK sources: England Approved Documents A and C; BS EN 1990/1991/1992/1993/1995/1996/1997 with their applicable UK National Annexes. This trial uses first-generation assumptions; appointed designers must confirm editions and national choices. The Swedish Wood formulas are Eurocode educational guidance, not UK project approval.

AD-A / England Approved Document A

AD-C / England Approved Document C

SCI-BASIS / SCI / BCSA Eurocodes and UK National Annexes

SCI-MEMBER / SCI / BCSA member design

BS-UB / British Steel section properties

JJI / James Jones JJI technical manual - Table 16, p28

CATNIC / Catnic CG150/100 datasheet

BRICK / Wienerberger UK standard brick dimensions

BLOCK / Forterra Conbloc aggregate block guide

SCIA / SCIA Eurocode steel classification examples

UK-STUD / Planning Portal: internal-wall structural support

EC5-TIMBER / Swedish Wood / TDUK Eurocode 5 material data and formulas (national choices require UK confirmation)

UK-C24 / MGM Timber UK: finished C24 stock sizes

UK-LVL / Metsa Wood UK: Kerto LVL load-bearing studs

Audit: trials.json contains all inputs, rejected and passing limited screens, supplier data references and source hashes. Rebuilds reject stale source files. Unit checks cover equilibrium/units, supplier-table boundaries, unsupported load ratios, creep and independently integrated earth/water pressure.

Appendix: the following P01 coordination sheets show current architectural structural zones, not fully designed members. Trial directions/header/stud axes are also available as an optional layer in the local structural viewer. No approved structural design or changed architectural model is implied.

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Design basis and route to construction

Current Proposed model / brick cavity masonry / concrete basement concept

What this issue establishes

An interactive structural register linked to the current architectural model. The same element geometries generate the 3D workspace, floor plans, projected elevations and section. It identifies envelope zones, modelled opening heads and named support studies.

Owner decisions

New house walls: brick outer leaf, insulated cavity and load-bearing blockwork inner leaf. The separate garden workshop retains its 215 mm solid-masonry concept. Ground and engineering reports are not available, confirmed by the owner on 26 October 2026.

Known architectural constraints

Current basement footprint: 115.525 m2, with its stepped courtyard edge. New wing roof: continuous, matching 29.745-degree pitches. Loft-only courtyard projection: 720 mm. The accepted original-house stair has approximately 2.01 m checked architectural headroom in places; actual structural depth and tolerances remain critical.

Dimensions and quantity meaning

Dimensions describe model envelopes, not surveyed spans or fabrication lengths. Areas are projected architectural geometry. No whole-house brick, block, steel or concrete procurement quantities have been issued. Model fragments must first be reconciled into wall leaves, structural floor systems and designed foundations.

What is not designed

Steel sections and connections; joist type, size, spacing and direction; masonry strengths and bearings; lintel products and padstones; lateral restraint and stability; foundations; basement reinforcement and waterproofing joints; temporary works and excavation sequence.

Evidence required next

Commission a measured and structural survey, opening-up of existing structure and foundation investigations, geotechnical investigation with groundwater information, drainage/service records and tree constraints. The project structural engineer must set the investigation scope; trial pits and excavation need planned safe execution.

Issue path

1. Reconcile measured geometry and decide the wall/floor assemblies.
2. Agree vertical and lateral load paths and support locations.
3. Calculate and check members, connections and foundation reactions.
4. Coordinate basement structure, waterproofing and temporary works.
5. Add the checked design to this register and produce construction drawings, reinforcement and member schedules.
6. Complete the applicable building-control process and site verification before release for construction.

Current approval

Zero construction-approved elements. A competent structural engineer with basement experience must design/check and take responsibility for the structural scheme. These drawings do not establish safe excavation or a compliant structural solution.

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Calculation and drawing deliverables

Checklist for the appointed structural engineer and coordinated model

Loads and stability

Site-specific wind and snow; permanent, imposed, partition and point loads; terraces, plant and vehicle effects where applicable. Confirm current standards/UK National Annexes and design combinations. Define diaphragms, restraint, racking frames and the complete reaction path to the ground.

Steel and opening supports

Each final member reference must link to load derivation, analysis, bending/shear/buckling checks, deflection criteria, connection design, bearings/padstones and reactions. Coordinate cavity outer-leaf support and thermal/fire/corrosion requirements. Existing openings require verification, not automatic replacement.

Floors and roof

Choose systems after support layout: timber/I-joist/metal-web or other engineered floors as appropriate; no joist direction has been selected. Provide spans, spacing, trimming, hangers, bearing, restraint, vibration and service-zone details. Coordinate ridge/hip/dormer members, rooflights and cantilever backspan.

Basement and foundations

Select the foundation/retaining system from geotechnical results and adjoining foundation surveys. Check earth/water/surcharge pressures, stability, settlement, flotation, bending, shear, punching and cracking as applicable. Issue concrete specifications, cover, reinforcement plans/sections, bar schedules, joints and waterstops, coordinated with the waterproofing specialist.

Masonry and quantity take-off

Final drawings must separate brick outer leaf, cavity/insulation and block inner leaf. Include verified openings, lintels, cavity trays, closers, ties, restraint, movement joints, DPC, substructure and local bearing details. Quantify each material from coordinated assemblies, with explicit deductions, specials and waste; do not sum mesh-envelope areas.

Construction sequence and record

Define temporary support, excavation and retaining-wall sequence, monitoring and adjoining-building protection with the designers/contractor. Record inspections and design changes. Fabrication and procurement dimensions must be checked on site. No sequence or reinforcement schedule is supplied in this P01 issue.

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Key coordination questions

Items to resolve while protecting the accepted layout

Q01 / House wall build-up

The visual model contains thin single wall shells. Reconcile brick/cavity/blockwork thickness before freezing room sizes or ordering lintels.

Q02 / Basement and existing foundations

Investigate soil, groundwater, drains, tree influence and adjacent foundations. No safe excavation, retaining-wall thickness, footing size or reinforcement is established.

Q03 / Large openings and transfer loads

Glazed courtyards, garage openings and the chamfered garden room require support and stability design. A window frame is not assumed to support the floor or masonry above.

Q04 / Continuous roof and 720 mm projection

Coordinate ridge/hip/dormer support and cantilever backspan with a structural engineer while retaining the accepted flat-ceiling route.

Q05 / Floor depth and stair headroom

The checked architectural headroom is only about 2.01 m in places. Engineered floor depths, finishes and tolerances may require layout adjustment.

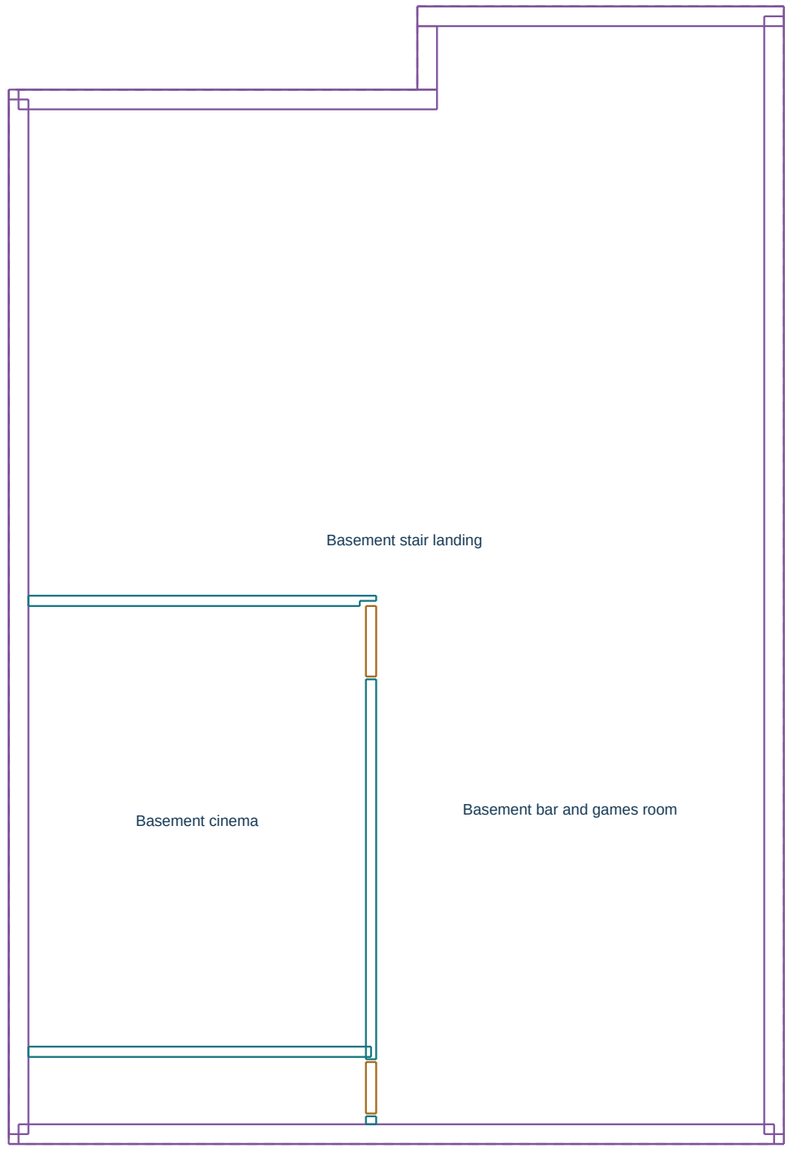
Q06 / Quantities and completeness

Model opening heads are candidates, not a complete lintel order. Whole-house brick/block quantities await leaf geometry, verified openings, movement joints, below-ground work and waste allowances.

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Basement coordination plan

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



5 m · model geometry

Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Plans cut at 1.20 m above the selected floor. Floor outlines and opening heads are projected for coordination. Hidden/interior lines remain visible.

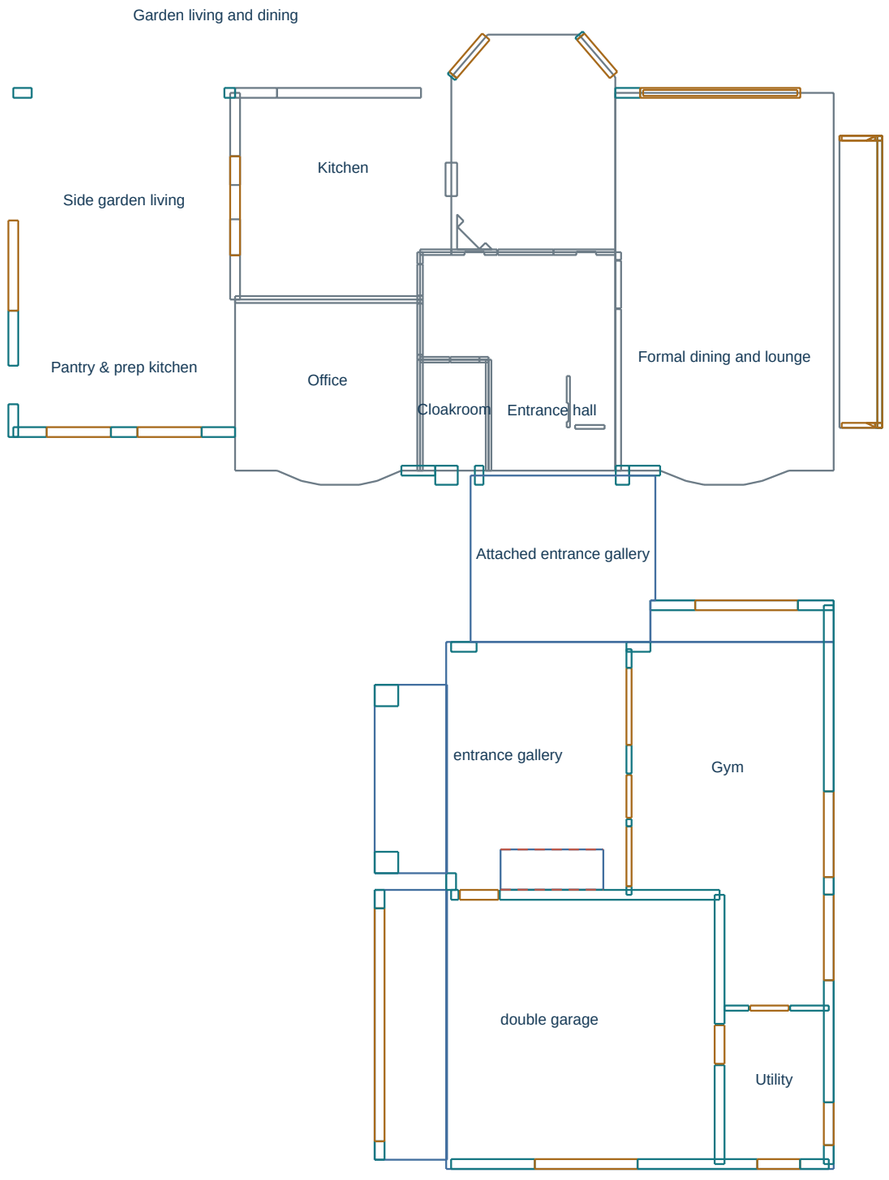
No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

P01 - COORDINATION ONLY - NOT FOR CONSTRUCTION

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Ground floor coordination plan

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



5 m · model geometry

Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Plans cut at 1.20 m above the selected floor. Floor outlines and opening heads are projected for coordination. Hidden/interior lines remain visible.

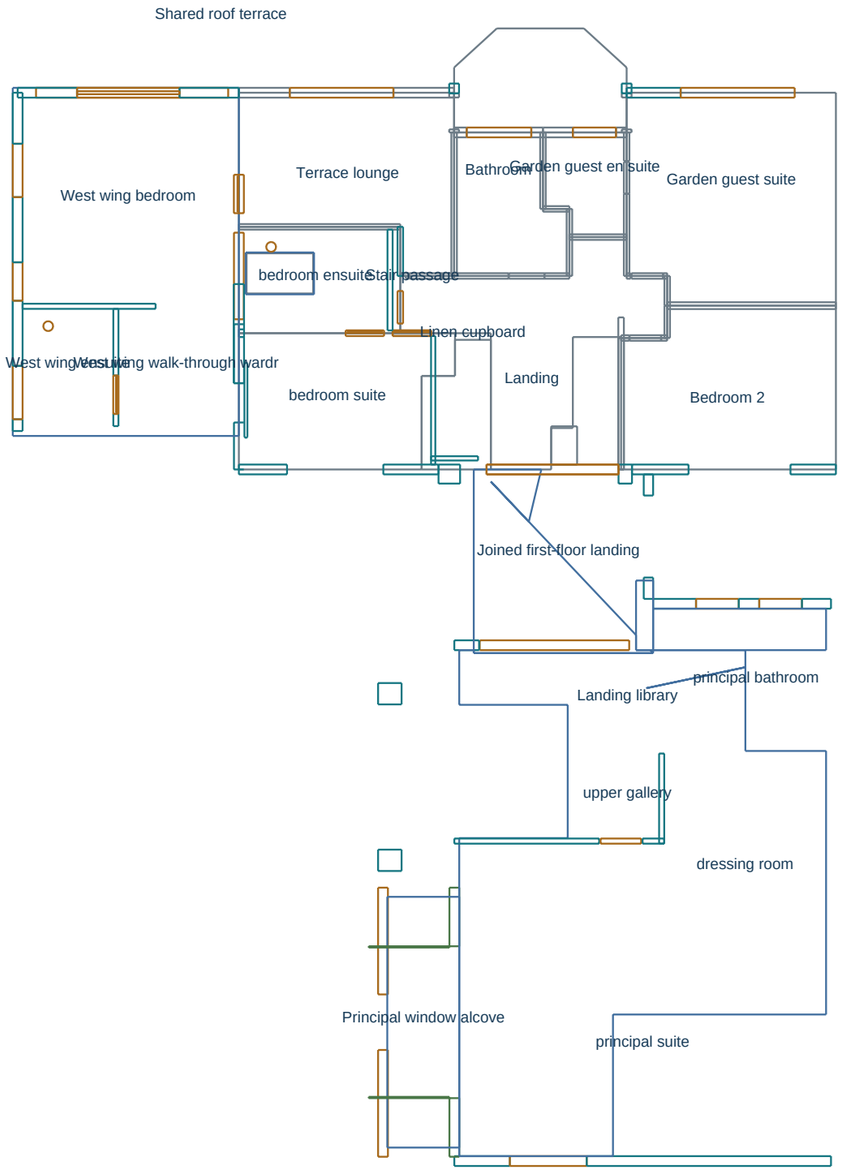
No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

P01 - COORDINATION ONLY - NOT FOR CONSTRUCTION

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

First floor coordination plan

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



5 m · model geometry

Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Plans cut at 1.20 m above the selected floor. Floor outlines and opening heads are projected for coordination. Hidden/interior lines remain visible.

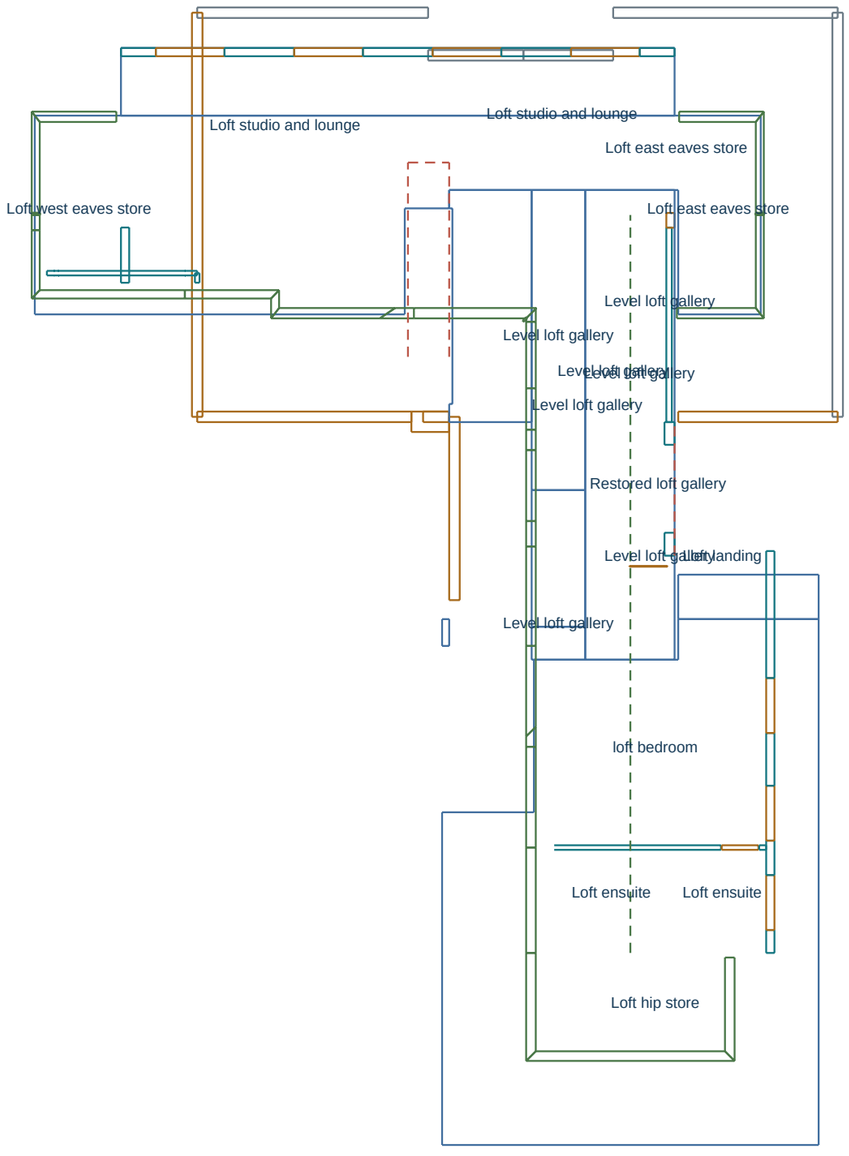
No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

P01 - COORDINATION ONLY - NOT FOR CONSTRUCTION

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Loft coordination plan

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



5 m · model geometry

Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Plans cut at 1.20 m above the selected floor. Floor outlines and opening heads are projected for coordination. Hidden/interior lines remain visible.

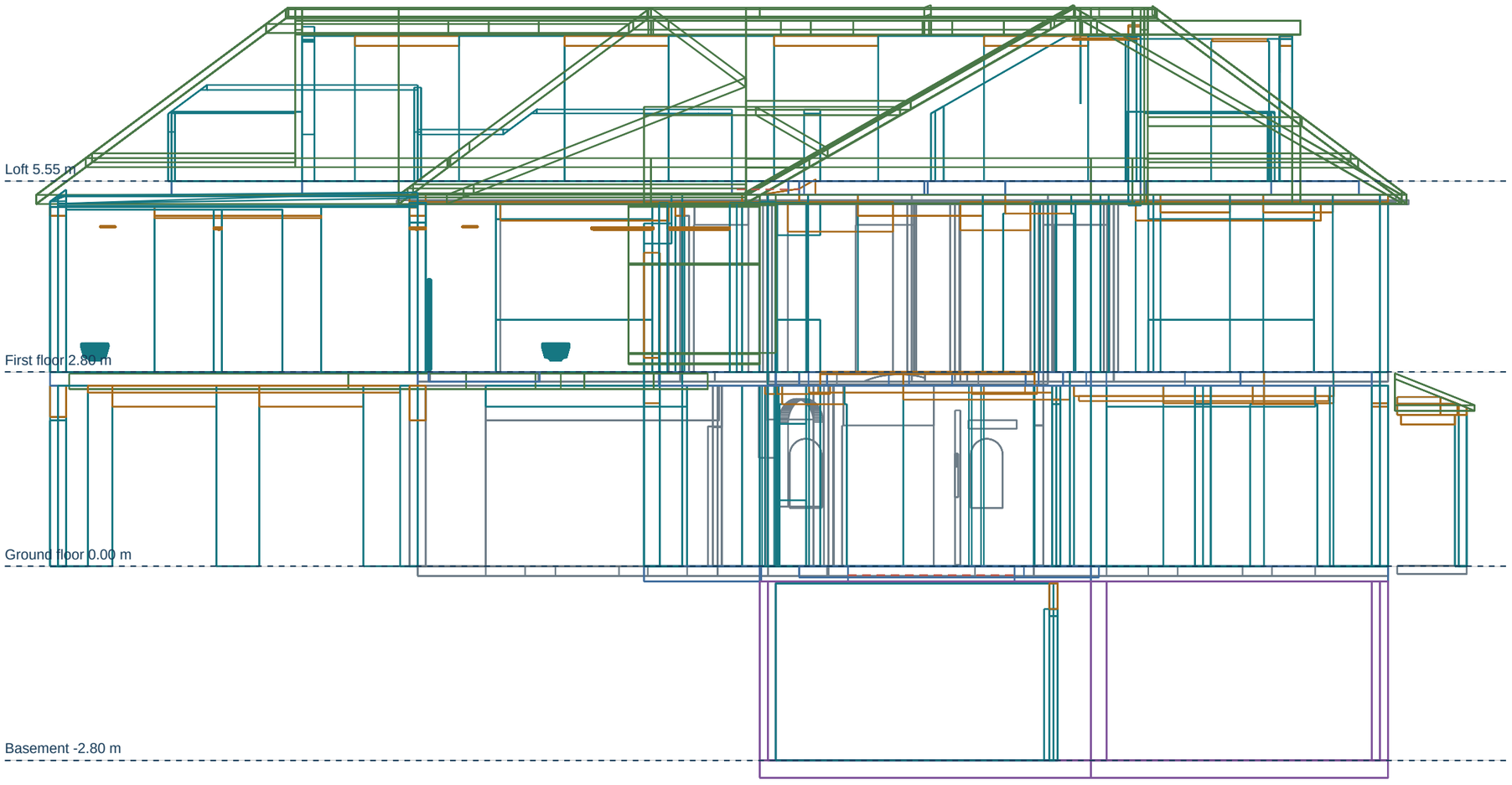
No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

P01 - COORDINATION ONLY - NOT FOR CONSTRUCTION

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Front structural projection

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



P01 - COORDINATION ONLY - NOT FOR CONSTRUCTION

Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

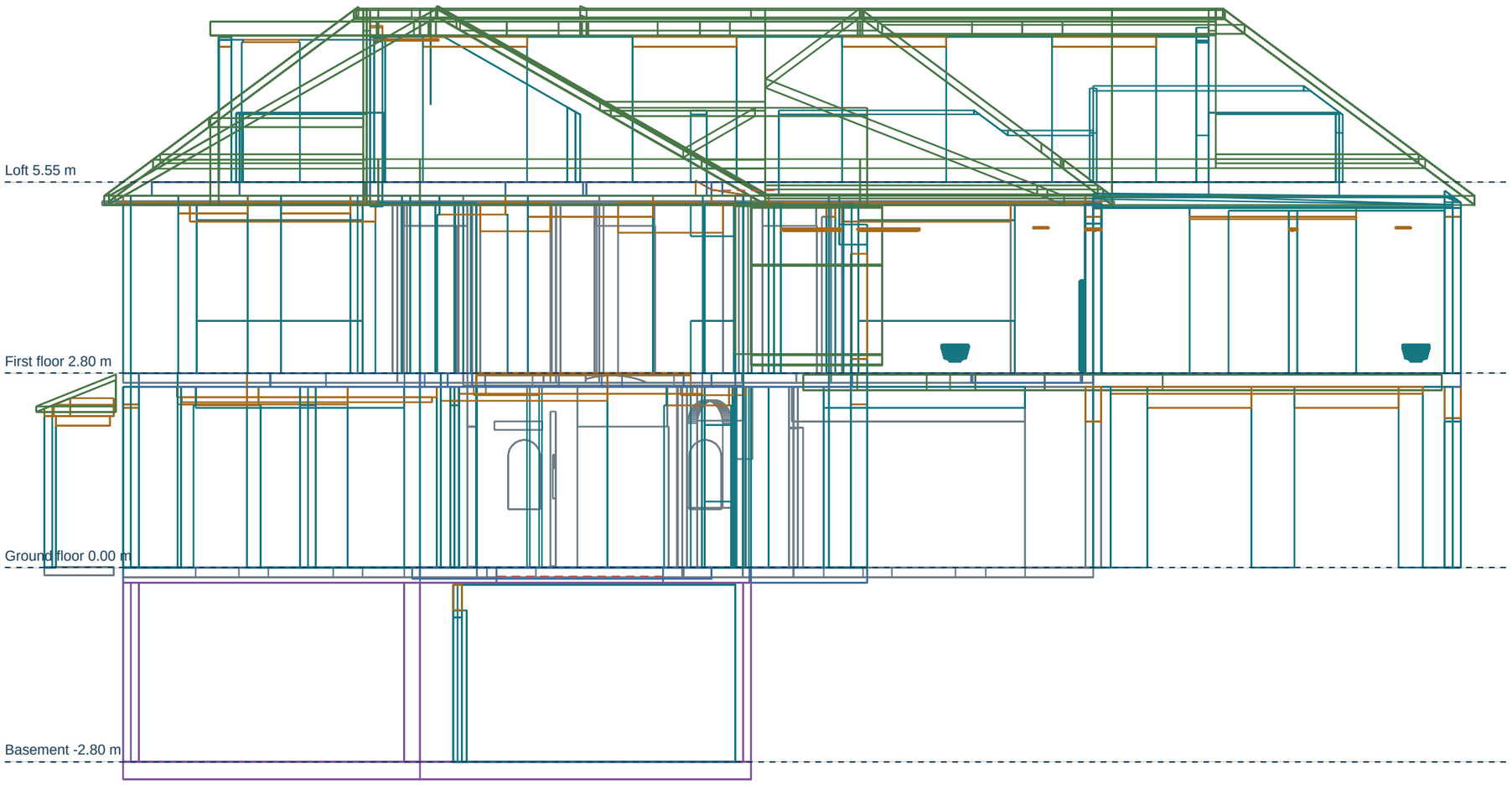
This is an orthographic structural projection with internal lines visible, not a hidden-line architectural elevation.

No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Rear structural projection

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

This is an orthographic structural projection with internal lines visible, not a hidden-line architectural elevation.

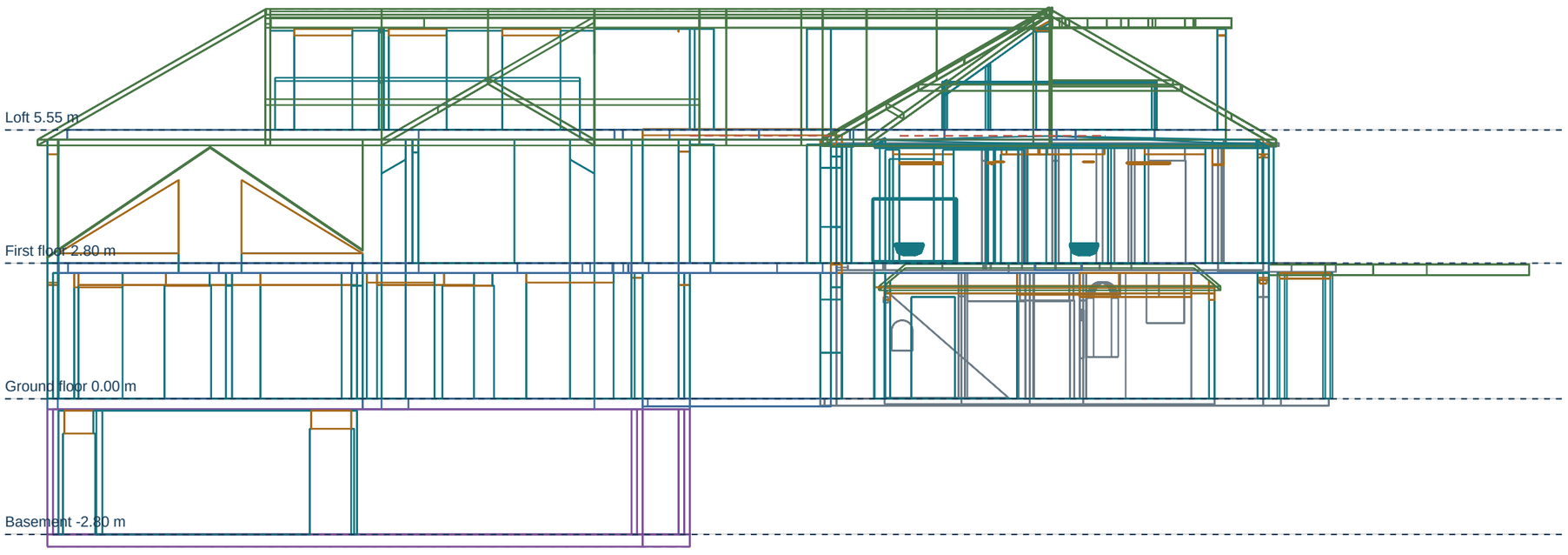
No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

P01 - COORDINATION ONLY - NOT FOR CONSTRUCTION

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

West structural projection

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



5 m · model geometry

P01 - COORDINATION ONLY - NOT FOR CONSTRUCTION

Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

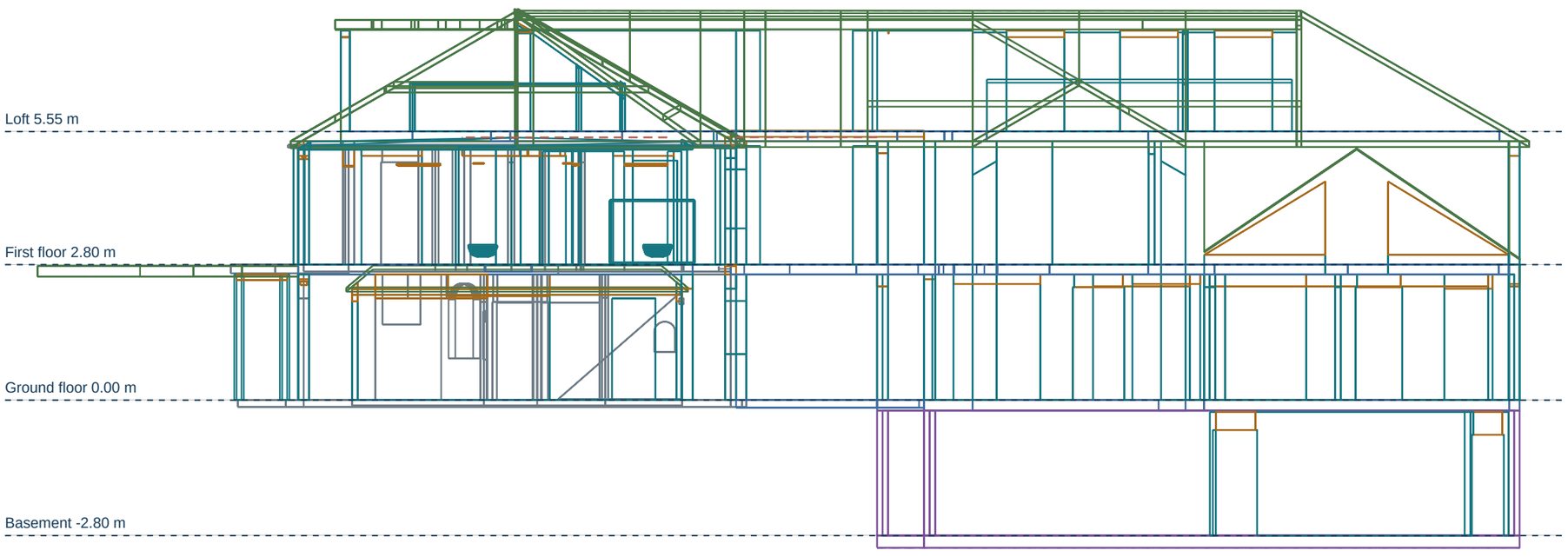
This is an orthographic structural projection with internal lines visible, not a hidden-line architectural elevation.

No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

East structural projection

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



5 m · model geometry

Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

This is an orthographic structural projection with internal lines visible, not a hidden-line architectural elevation.

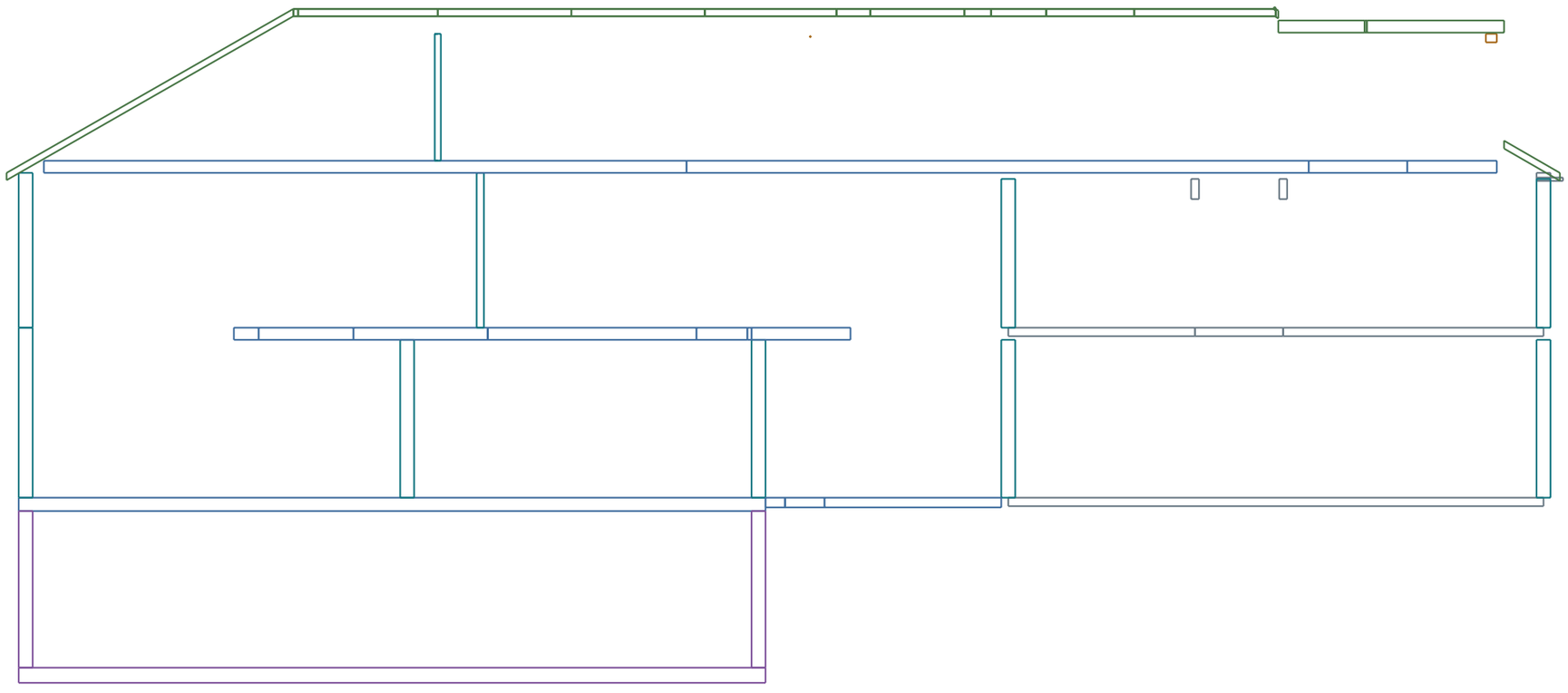
No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

P01 - COORDINATION ONLY - NOT FOR CONSTRUCTION

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Longitudinal coordination section

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



5 m · model geometry

Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Exact mesh cut at model X = 9.455 m. No below-ground foundation geometry beyond the architectural basement base is inferred.

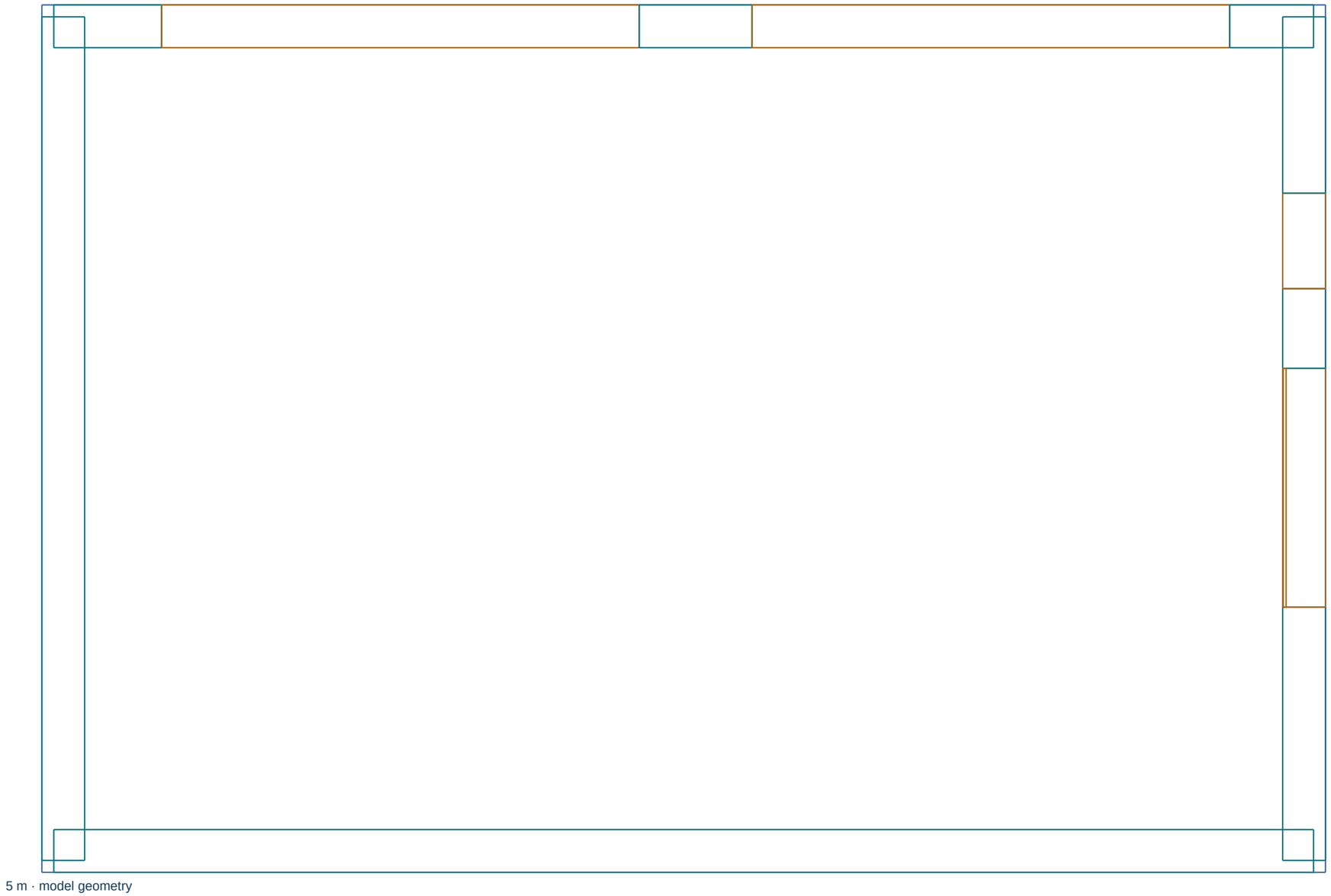
No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

P01 - COORDINATION ONLY - NOT FOR CONSTRUCTION

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Garden workshop coordination plan

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Plans cut at 1.20 m above the selected floor. Floor outlines and opening heads are projected for coordination. Hidden/interior lines remain visible.

No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

Evidence and reference register

Source geometry, issue identity and independent guidance

Model source

outputs/output-proposed-compact/geometry.json
outputs/output-proposed-compact/navigation.json
These are saved architectural exports. They are not a structural measured survey.

Source fingerprint

Geometry SHA-256:
3fb81b911061a35f7b4903f8f7d867fd
5aeafc029de295dd7e00dc152f7b87b7

Live element register

The local workspace is available at http://127.0.0.1:8765/structure/. Its model.json and schedule.json retain complete element references and source hashes. Geometry changes invalidate the structural build rather than silently leaving a stale register.

England - structural guidance

Approved Document A: structure
Foundations, walls, floors and roofs; checked 26 October 2026.

Ground and moisture

Approved Document C
Ground conditions and moisture guidance; checked 26 October 2026.

Domestic structural engineering

IStructE: introduction for domestic clients
Structural assessment, calculations, foundation investigation and on-site measurement; checked 26 October 2026.

Scope of this issue

Main house and garden workshop architectural bodies plus named support studies. Landscape, pool, pavilion, decorative layers and services excluded. Modelled head/lintel candidates only; not every structural opening has been reconciled.